

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

May 2026 – Present

Undergraduate Research Assistant

East Lansing, MI

- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.
- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, integrating nav2's MPPI controller rather than hand-rolling one to keep the C++ work on security, not motion control.
- Reproduced a published adversarial-robotics attack loop (attacker, target, judge, syntax-checker) as the baseline, then drove the decomposition attack to an 80% success rate against the per-step triage gate.

Robostangs (FRC Team 548)

Jun 2023 – May 2024

Software Captain

Northville, MI

- Delivered the team's first multi-piece autonomous routine, chaining vision-corrected driving paths with timed intake-and-shoot actions so the robot scored multiple game pieces with no driver input during the autonomous period.
- Anchored auto-aim on real-time, vision-corrected localization that fused two AprilTag cameras with swerve odometry on a dedicated 250Hz Java thread with latency compensation, holding pose accurate during aggressive motion.
- Hardened pose estimation against bad vision data by gating camera measurements on tag count, target area, and angular velocity before fusing them, rejecting estimates that would otherwise poison the localization filter.

PROJECTS

WizViz | *Pose Inference, Gesture Classification*

devpost.com/software/wizviz

Webcam fighting game driven by real-time body-gesture recognition, winner of the Interactive Media track at SpartaHack X

- Stabilized noisy per-frame pose output by accumulating every frame's gesture classification across the turn window and taking the dominant sustained gesture, instead of reading a single jittery frame at the turn boundary.
- Classified five distinct combat gestures from body landmarks with a scale-invariant geometric model normalized by torso length, making detection work across players and camera distances without any labeled training data.
- Ran pose inference asynchronously off the render loop with a monotonic timestamp guard that drops out-of-order results, sustaining 60 FPS with sub-20ms input lag instead of blocking on per-frame detection.

Dadei | *LLM Approval Gate, Speaker ID*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Routed all model-proposed side effects through one approval chokepoint with a cancelable countdown, giving every irreversible action one auditable seam between the LLM and the outside world.
- Engineered a no-merge-first speaker-ID model using an EMA voice centroid plus a bounded prototype bank with floor-and-margin gating, abstaining into a new identity on ambiguity, attributing 96% of utterances to the correct speaker.

fliks | *ML Rating Model*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Architected an ML rating service that grades each clip independently and shows its verdict beside the crowd's, with those crowdsourced ratings serving as the labeled data the classifier trains on.

TECHNICAL SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, Bash

AI / ML PyTorch

Robotics ROS2, Gazebo, AprilTag

Tools Git, Linux, ROSBag